

Arjun Giriraj Day 09

1. What is Linux?

⇒ Linux is an operating system used in servers, cloud systems, networking devices, cybersecurity environment, and Android based system.

2. What is a Linux distribution?

⇒ Linux distribution is a distribution packages which contains linux kernel, system utilities, a package manager, applications, desktop environment.

3. Name four Linux distributions.

⇒ Ubuntu, Kali linux, Fedora, Red hat.

4. What is Kali Linux?

⇒ Kali linux is a linux based operating system designed for security testing and cybersecurity works.

5. What is a terminal?

⇒ Terminal lets you use or control the linux using commands.

6. What does the whoami command do?

⇒ Returns the current username.

7. What does hostname display?

⇒ Shows the computers hostname.

8. What does pwd mean?

⇒ Present working directory.

9. What is the purpose of ls?

⇒ To list the files and directory in the working directory.

10. Explain the difference between ls, ls -l and ls -a.

⇒ ls -l returns the details listing with permission, size, owner, date.

⇒ ls -a returns the all files the directory including hidden files.

11. What does cd do?

⇒ It changes the directory.

12. What does cd .. do?

⇒ cd .. return to the parent directory.

13. What does cd ~ do?

⇒ cd ~ returns the user to the home directory.

14. What does cd / do?

⇒ `cd /` returns the user to the filesystem root directory.

15. What is the Linux root directory?

⇒ The parent directory of the linux filesystem.

16. What is the difference between the root directory and the root user?

⇒ Root directory is on the top of the tree in the directory tree structure. The parent directory of the linux file system.

⇒ Root user is the administrative user of the filesystem which have all rights.

17. What is an absolute path?

⇒ It always start with the filesystem root.

18. What is a relative path?

⇒ shows the parent directory of the current file.

19. How are hidden Linux files commonly identified?

⇒ It name begin with the dot .

20. Why are Linux skills important for cybersecurity?

⇒ Cybersercurity professionals use linux to inspect files, navigate log files, run security tools, analyze network. Before using advanced linux tool user must be comfortable with the basic linux navigation.

PRACTICAL SCREENSHOTS (from your own Linux system/VM):

`whoami · hostname · pwd · ls · ls -l · ls -a · cd / · cd ~`

```
venom@VENOM:/$ ls
bin  etc  initrd.img.old  lib64  mnt  proc  rxdenv  sys  usr  vmlinuz.old
boot home  lib          lost+found  opt  root  sbin  System  var  vmlinuz
dev  initrd.img  lib32  media  path  run  srv  tmp  vmlinuz

venom@VENOM:/$ ls -l
total 88
drwxr-xr-x 1 root root    7 Aug  8 2025 bin -> usr/bin
drwxr-xr-x 4 root root   4096 Aug 27 14:38 boot
drwxr-xr-x 20 root root  3860 Sep 14 13:19 dev
drwxr-xr-x 227 root root 20480 Sep 12 18:10 etc
drwxr-xr-x 3 root root   4096 Aug  8 2025 home
lrwxrwxrwx 1 root root    32 Aug 27 14:34 initrd.img -> boot/initrd.img-7.1.5+kali-amd64
lrwxrwxrwx 1 root root    34 Aug 27 14:34 initrd.img.old -> boot/initrd.img-6.18.12+kali-amd64
drwxr-xr-x 1 root root    7 Aug  8 2025 lib -> usr/lib
lrwxrwxrwx 1 root root    9 Aug  8 2025 lib32 -> usr/lib32
lrwxrwxrwx 1 root root    9 Aug  8 2025 lib64 -> usr/lib64
drwx----- 2 root root 16384 Aug  8 2025 lost+found
drwxr-xr-x 4 root root   4096 Sep  8 2025 media
drwxr-xr-x 2 root root   4096 Aug  8 2025 mnt
drwxr-xr-x 6 root root   4096 Mar 24 18:36 opt
drwxr-xr-x 3 root root   4096 Dec 25 2025 path
dr-xr-xr-x 315 root root    0 Sep 14 13:19 proc
drwx----- 15 root root   4096 Aug 27 14:36 root
drwxr-xr-x 46 root root  1040 Sep 14 13:56 run
drwxrwxr-x 3 venom venom 4096 Jan 22 2026 rxdenv
lrwxrwxrwx 1 root root    8 Aug  8 2025 sbin -> usr/sbin
drwxr-xr-x 3 root root   4096 Aug  8 2025 srv
dr-xr-xr-x 13 root root    0 Sep 14 13:19 sys
drwxr-xr-x 3 root root   4096 May 22 15:24 System
drwxrwxrwt 18 root root   420 Sep 14 15:31 tmp
drwxr-xr-x 17 root root   4096 Aug  8 2025 usr
drwxr-xr-x 12 root root   4096 Aug  8 2025 var
lrwxrwxrwx 1 root root    29 Aug 27 14:34 vmlinuz -> boot/vmlinuz-7.1.5+kali-amd64
lrwxrwxrwx 1 root root    31 Aug 27 14:34 vmlinuz.old -> boot/vmlinuz-6.18.12+kali-amd64

venom@VENOM:/$ ls -a
.  ..  bin  boot  .cache  dev  etc  home  initrd.img  initrd.img.old  lib  lib32  lib64  lost+found  media  mnt  opt  path  proc  root  run  rxdenv  sbin  srv  sys  System  tmp  usr  var  vmlinuz  vmlinuz.old

venom@VENOM:/$ hostname
VENOM

venom@VENOM:/$ whoami
venom

venom@VENOM:/$ cd /
venom@VENOM:/$ cd ~
venom@VENOM:~/$ cd Desktop
venom@VENOM:~/Desktop/$
```